



Yakeen Plus (2025)

PRACTICE TEST- 12

DURATION :200 Minutes

DATE : 04/12/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (2)
2. (3)
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4. (4)
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CHEMISTRY

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150. (4)

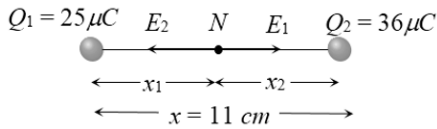
ZOOLOGY

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198. (3)
199. (3)
200. (3)

SECTION-(I) PHYSICS

1. (2)

The schematic diagram of distribution of charges on x-axis is shown in figure below:



Total force acting on 1 C charge is given by

$$F = \frac{1}{4\pi\epsilon_0} \left[\frac{1 \times 1 \times 10^{-6}}{(1)^2} + \frac{1 \times 1 \times 10^{-6}}{(2)^2} + \frac{1 \times 1 \times 10^{-6}}{(4)^2} + \frac{1 \times 1 \times 10^{-6}}{(8)^2} + \dots \infty \right]$$

$$= \frac{10^{-6}}{4\pi\epsilon_0} \left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \infty \right)$$

$$= 9 \times 10^9 \times 10^{-6} \left(\frac{1}{1 - \frac{1}{4}} \right)$$

$$= 9 \times 10^9 \times 10^{-6} \times \frac{4}{3} = 9 \times 10^3 \times \frac{4}{3} = 12000 \text{ N}$$

[New NCERT Class 11th Page No. 11]

2. (3)

Electric field at the centre of arc

$$E = \frac{2k\lambda}{r} \sin\left(\frac{\theta}{2}\right)$$

For half ring $\theta = \pi$

$$E = \frac{2\lambda}{4\pi\epsilon_0 a} \sin\left(\frac{\pi}{2}\right) = \frac{\lambda}{2\pi\epsilon_0 a}$$

[New NCERT Class 12th Page No. 28]

3. (2)

Here $F = \frac{KQQ}{r^2}$ (attractive)

Now when charge is transferred A has charge

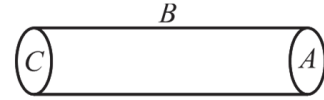
$$\frac{3Q}{4}, \text{ and B has charge } \frac{3Q}{4}$$

$$F' = \frac{K\left(\frac{3Q}{4}\right)\left(\frac{3Q}{4}\right)}{r^2}$$

$$= \frac{9}{16} \frac{KQ^2}{r^2} = \frac{9F}{16}$$

[New NCERT Class 12th Page No. 28]

4. (4)



$$\text{Net flux through the all surface} = \frac{Q}{\epsilon_0}$$

$$\text{Flux through curved surface} = \phi$$

Hence, flux through plane surface

$$A = \frac{1}{2} \left(\frac{Q}{\epsilon_0} - \phi \right)$$

5. (1)

Acceleration of a charged particle of charge q and mass m moving in a uniform electric field of strength E is $a = \frac{F}{m} = \frac{qE}{m}$

6. (3)

If electrostatic field lines formed closed loops, then work done in taking a charge along this loop will be non-zero which contradicts the fact that electrostatic force is conservative. Therefore electrostatic field lines can never be closed. Thus, both assertion and reason are correct and reason is the correct explanation of assertion.

7. (1)

Let q be the charge on each drop of radius r .
Energy before merger

$$U_1 = 1000 \times \frac{q^2}{2C} = 10^3 \frac{q^2}{2C}$$

On merger,

$$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3, \therefore R = 10r$$

Capacity becomes 10 times.

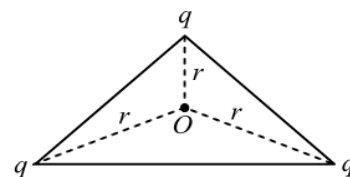
Energy after merger

$$= U_2 = \frac{(1000q)^2}{2(10C)} = 10^5 \frac{q^2}{2C}$$

$$\therefore U_2 / U_1 = 100$$

[New NCERT Class 12th Page No. 56]

8. (3)



$$V = k \frac{Q}{r} \times 3 \neq 0$$

But $E = 0$

[New NCERT Class 12th Page No. 52]

9. (4)

$$i_{AO} + i_{CO} = i_{OB}$$

$$\frac{20 - V_0}{10} + \frac{30 - V_0}{10} = \frac{V_0 - 10}{10}$$

$$V_0 = 20V$$

$$i_{AO} = \frac{20 - V_0}{10} = 0$$

10. (3)

$$v_d \propto \frac{1}{A} \text{ for } i = \text{constant}$$

[New NCERT Class 12th Page No. 96]

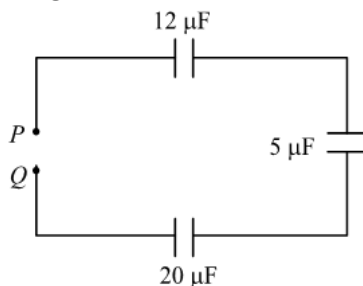
11. (2)

$$V_{AB} = V_A - V_B = -3 \times 3 + 3 \times 6 = 9V$$

[New NCERT Class 12th Page No. 119]

12. (2)

The given circuit can be drawn as



$$\text{where } C = (3 + 2)\mu F = 5\mu F$$

$$\frac{1}{C_{PQ}} = \frac{1}{5} + \frac{1}{20} + \frac{1}{12} = \frac{20}{60} = \frac{1}{3}$$

$$\Rightarrow C_{PQ} = 3\mu F$$

[New NCERT Class 12th Page No. 71, 72]

13. (4)

$$V_A - 4 - 3 - 2 - V_B = 0$$

$$V_A - V_B = 9V$$

14. (3)

$$E = \frac{E_0}{K}$$

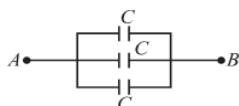
$$\therefore K_1 < K_2$$

$$\text{Then, } E_1 > E_2$$

$$\text{Finally, } E_0 > E_1 > E_2$$

[New NCERT Class 12th Page No. 69, 70]

15. (2)



$$C_{AB} = 3C$$

[New NCERT Class 12th Page No. 71, 72]

16. (2)

$$r = \frac{P}{qB}$$

$$\text{So, } r_d : r_\alpha = 3:1$$

[New NCERT Class 12th Page No. 138]

17. (2)

$$\int \vec{B} \cdot d\vec{l} = \mu_0 l_{\text{inside}} = \mu_0 (4i - i) = 3\mu_0 i$$

[New NCERT Class 12th Page No. 147]

18. (3)

Since F is always perpendicular to velocity. Hence speed remains constant.

[New NCERT Class 12th Page No. 138]

19. (2)

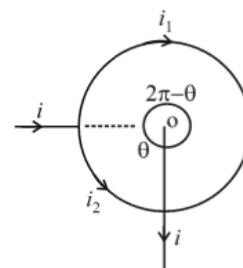
$$l_{in} = 0, \oint \vec{B} \cdot d\vec{l} = \mu_0 i_{in} = 0$$

[New NCERT Class 12th Page No. 147]

20. (2)

Net magnetic field at point 'P'

$$B_{net} = \vec{B}_1 + \vec{B}_2$$



Here $\vec{B}_1$ and $\vec{B}_2$ are equal in magnitude and opposite in direction.

$$\text{Hence, } B_{net} = B_1 - B_2$$

$$B_1 = \frac{\mu_0 i_1}{2R} \left(\frac{2\pi - \theta}{2\pi} \right)$$

$$B_2 = \frac{\mu_0 i_1}{2R} \left(\frac{\theta}{2\pi} \right)$$

$$\text{and } i_2 = \frac{3i}{4}, i_1 = \frac{i}{4}$$

[New NCERT Class 12th Page No. 145]

21. (2)

$$B = \frac{\mu_0 I}{2r},$$

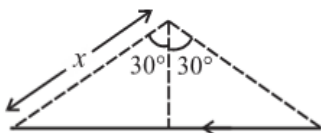
when made n turns radius becomes r'

$$n \times 2\pi r' = 2\pi r \Rightarrow r' = \frac{r}{n}$$

$$\text{Now, } B' = \frac{\mu_0 n I}{2r'} = n^2 \frac{\mu_0 I}{2r} = n^2 B$$

[New NCERT Class 12th Page No. 146]

22. (1)



$$B = \frac{\mu_0 I}{4\pi \cos 30^\circ} [\sin 30^\circ + \sin 30^\circ]$$

$$B = \frac{\mu_0 I}{4\pi \left(\frac{\sqrt{3}}{2}\right)} \left(\frac{1}{2} + \frac{1}{2}\right) = \frac{\mu_0 I}{2\sqrt{3}\pi x}$$

[New NCERT Class 12th Page No. 144]

23. (3)

Direction of magnetic fields at point A and C is opposite but having same magnitude so magnetic field at these points is zero.

[New NCERT Class 12th Page No. 119]

24. (2)

$$\phi = 6t^2 - 5t + 1$$

$$\varepsilon = -\frac{d\phi}{dt} = -12t + 5$$

$$I = \frac{\varepsilon}{R} = \frac{-12\left(\frac{1}{4}\right) + 5}{10} = 0.2 \text{ A}$$

[New NCERT Class 12th Page No. 180]

25. (1)

Electric field is produced by time varying magnetic flux.

[New NCERT Class 12th Page No. 182]

26. (1)

The magnetic susceptibility of a material is a measure of the ease with which a specimen of that material can be magnetised in a magnetising field. For a diamagnetic substance, magnetic susceptibility (χ_m) is independent of temperature.

[New NCERT Class 12th Page No. 182]

27. (2)

From Faraday's second law, emf induced in the circuit

$$e = \frac{\Delta\phi}{\Delta t}$$

If R is the resistance of the circuit, then

$$i = \frac{e}{R} = \frac{\Delta\phi}{R\Delta t}$$

Thus, charge passing through the circuit,

$$q = i \times \Delta t$$

$$\Rightarrow q = \frac{\Delta\phi}{R\Delta t} \cdot \Delta t$$

$$\Rightarrow q = \frac{\Delta\phi}{R}$$

[New NCERT Class 12th Page No. 190]

28. (4)

$$\text{At resonance } I = \frac{V}{R}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

and voltage across resistor, inductor and capacitor will be in different phase.

$$I = \frac{200}{20}$$

$$I = 10 \text{ A}$$

$$f = \frac{1}{2\pi\sqrt{\frac{2}{\pi} \times 10^{-6} \times \frac{2}{\pi}}} = \frac{1000}{4} = 250 \text{ Hz}$$

[New NCERT Class 12th Page No. 187 to 189]

29. (4)

$$\begin{aligned} V_A &= \sqrt{V_R^2 + (V_L - V_C)^2} \\ &= \sqrt{(60)^2 + (120 - 40)^2} \\ &= \sqrt{60^2 + (80)^2} = 100 \text{ V} \end{aligned}$$

[New NCERT Class 12th Page No. 187 to 189]

30. (2)

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

[New NCERT Class 12th Page No. 187 to 189]

31. (2)

We know that quality factor (Q),

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$Q^2 = \frac{L}{R^2 C} \text{ or } C = \frac{L}{Q^2 R^2}$$

[New NCERT Class 12th Page No. 187 to 189]

32. (4)

$$\sqrt{\frac{V_0^2 + 0^2}{2}} = \frac{V_0}{\sqrt{2}}$$

[New NCERT Class 12th Page No. 178, 179]

33. (3)

$$20 = 100\pi L$$

$$\frac{1}{5\pi} = L$$

$$X_L = \omega L = 2\pi(100)\left(\frac{1}{5\pi}\right) = 40\Omega$$

$$\text{and } R = 30\Omega$$

$$\Rightarrow Z = \sqrt{R^2 + X_L^2} = \sqrt{(30)^2 + (40)^2} = 50\Omega$$

$$\text{So, } I = \frac{200}{50} = 4A$$

[New NCERT Class 12th Page No. 187 to 189]

34. (2)

Here current, $i = i_1 \sin \omega t + i_2 \cos \omega t$

$$\Rightarrow i = \sqrt{i_1^2 + i_2^2} \left[\frac{i_2 \cos \omega t}{\sqrt{i_1^2 + i_2^2}} + \frac{i_1 \sin \omega t}{\sqrt{i_1^2 + i_2^2}} \right]$$

$$\Rightarrow i = \sqrt{i_1^2 + i_2^2} [\sin \alpha \cos \omega t + \cos \alpha \sin \omega t]$$

$$\Rightarrow i = \sqrt{i_1^2 + i_2^2} \sin(\omega t + \alpha) = i_0 \sin(\omega t + \alpha)$$

$$\Rightarrow i_{rms} = \frac{i_0}{\sqrt{2}} = \sqrt{\frac{i_1^2 + i_2^2}{2}}$$

[New NCERT Class 12th Page No. 187 to 189]

35. (3)

Resonance frequency in radian/second is

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 0.5 \times 10^{-6}}} = 500 \text{ rad/sec}$$

[New NCERT Class 12th Page No. 187 to 189]

36. (3)

Given electric and magnetic field energy is equal

$$\frac{q^2}{2C} = \frac{1}{2} LI^2 \quad \dots (i)$$

When the capacitor is completely charged,

$$\text{the total energy in the } LC = \frac{Q^2}{2C} \quad \dots (ii)$$

By (i) and (ii)

$$\text{Total energy} \Rightarrow \frac{Q^2}{2C} = \frac{q^2}{2C} + \frac{1}{2} LI^2$$

$$\Rightarrow \frac{Q^2}{2C} = \frac{q^2}{2C} + \frac{q^2}{2C}$$

$$q = \frac{Q}{\sqrt{2}}$$

[New NCERT Class 12th Page No. 187 to 189]

37. (4)

$$I = Q\omega \sin \omega t$$

$$I_{\max} = Q\omega$$

$$= \frac{60 \times 10^{-6} \times 100}{3 \times 10^{-4}} = 20 \text{ A}$$

OR

$$\frac{1}{2} LI_m^2 = \frac{1}{2} CV^2$$

$$LI_m^2 = CV^2$$

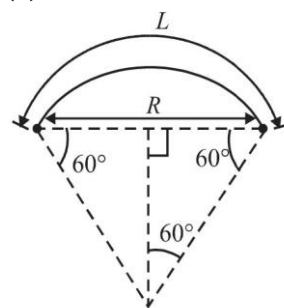
$$I_m^2 = \frac{CV^2}{L}$$

$$= \frac{60 \times 10^{-6} \times 100 \times 100}{1.5 \times 10^{-3}}$$

$$I_m = 20A$$

[New NCERT Class 12th Page No. 187 to 189]

38. (4)



$$M' = mR$$

$$\therefore L = \frac{\pi}{3} R$$

$$R = \frac{3L}{\pi}$$

$$\text{Now, } M' = \frac{3mL}{\pi}$$

$$= \frac{3M}{\pi}$$

[New NCERT Class 12th Page No. 137]

39. (2)

$$\begin{aligned}\vec{\tau} &= \vec{M} \times \vec{B} \\ &= 50\hat{i} \times (0.5\hat{i} + 3\hat{j}) \\ &= 150\hat{k} \text{ N-m}\end{aligned}$$

40. (4)

Area under B-H curve = energy lost + per cycle.

[New NCERT Class 12th Page No. 149]

41. (4)

“emf will be induced in such away that it will oppose the cause which has produced it”.

[New NCERT Class 12th Page No. 161]

42. (4)

$$e = B2Rv$$

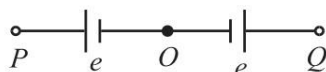
$$\therefore \pi R = L$$

$$R = \frac{L}{\pi}$$

$$e = \frac{2BLV}{\pi}$$

[New NCERT Class 12th Page No. 163]

43. (4)



$$\Delta v = V_P - V_\phi = 0$$

[New NCERT Class 12th Page No. 165]

44. (3)



$$\epsilon_1 = \frac{B\omega l^2}{2}$$

$$\text{and } \epsilon_2 = \frac{B\omega(3l)^2}{2}$$

$$\text{Now, } V_A - V_C = 4B\omega l^2$$

[New NCERT Class 12th Page No. 165]

45. (4)

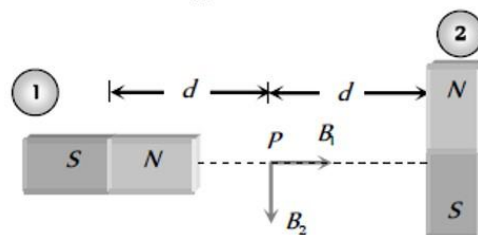
At point P net magnetic field

$$B_{\text{net}} = \sqrt{B_1^2 + B_2^2} \text{ where,}$$

$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3}$$

$$\text{and } B_2 = \frac{\mu_0}{4\pi} \cdot \frac{M}{d^3}$$

$$\Rightarrow B_{\text{net}} = \frac{\mu_0}{4\pi} \cdot \frac{\sqrt{5}M}{d^3}$$



[New NCERT Class 12th Page No. 149]

46. (4)

$$V = \frac{1}{\sqrt{\mu\epsilon}}$$

$$V = \frac{1}{\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}}$$

47. (3)

Electric charge at rest has electric field in the region around it, but no magnetic field. Whereas, a moving charge produces both the electric and magnetic fields. If a charge is moving with a constant velocity, then the electric and magnetic fields will not change with time, hence no EM wave will be produced. But, if the charge is moving with a non-zero acceleration, then both the electric and magnetic field will change with space and time, it then produces EM wave.

48. (1)

The magnetic field, $B_0 = E_0/c$,

where $c = 3 \times 10^8 \text{ ms}^{-1}$

$$B_0 = 6 / 3 \times 10^8 = 2 \times 10^{-8} \text{ T}$$

49. (3)

Gamma (γ)-rays has the shortest wavelength because it has higher frequency than UV-rays, microwaves and X-rays.

50. (4)

According to Maxwell, time varying electric field produced displacement current.

$$I_d = \epsilon_0 \frac{d\phi_E}{dt}$$

SECTION-(II) CHEMISTRY

51. (3)
Entropy: Solid < Liquid << Gas
[New NCERT Class 11th Page No. 159]

52. (4)
Internal energy depends upon the heat and work.
Internal energy is an intensive properties.
[New NCERT Class 11th Page No. 144]

53. (3)
 $\Delta S_{\text{system}} = nR \ln(V_2 / V_1) + nC_v \ln(T_2 / T_1)$
Isothermal process, so change in temperature is zero
 $\Delta S_{\text{system}} = nR \ln(V_2 / V_1)$
 $= 2 \times 8.314 \times \ln(100\text{L} / 10\text{L}) \text{ J/K}$
 $= 2 \times 8.314 \times 2.303 = 38.29 \text{ J/K}$
[New NCERT Class 11th Page No. 159]

54. (4)
The reaction for $\Delta_f H^\circ(\text{XY})$
 $\frac{1}{2} \text{X}_2(\text{g}) + \frac{1}{2} \text{Y}_2(\text{g}) \rightarrow \text{XY}(\text{g})$
Bond energies of X_2 , Y_2 and XY are x , $\frac{x}{2}$, x respectively.
 $\therefore \Delta H = \left(\frac{x}{2} + \frac{x}{4} \right) - X = -200$
On solving, we get
 $\Rightarrow \frac{-x}{4} = -200$
 $\Rightarrow x = 800 \text{ kJ/mole.}$
[New NCERT Class 11th Page No. 147]

55. (3)
 $\text{H}_2\text{O}_{(1)} \rightarrow \text{H}_2\text{O}_{(v)}$
5 5 mol
 $\Delta H = \Delta E + \Delta nRT$
 $9.72 \times 5 = \Delta E + 5 \times 2 \times 10^{-3} \times 373$
 $\Delta E = 48.6 - 3.73 = 44.87 \text{ kCal}$
[New NCERT Class 11th Page No. 143]

56. (3)
The number of unpaired electron in gaseous species of Mn^{3+} , Cr^{3+} and V^{3+} respectively are 4, 3, 2 respectively.
[New NCERT Class 12th Page No. 103]

57. (4)
The interstitial compounds are non-stoichiometric and melt at higher temperature than the metal and retain metallic conductivity.
[New NCERT Class 12th Page No. 104]

58. (3)
The rate of reaction doubles (increases by a factor of 2) with every 10°C rise in temperature. By increasing the temperature 40°C the rate of reaction becomes 16 times of initial rate.
[New NCERT Class 12th Page No. 78]

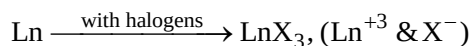
59. (4)
 $10^{15} e^{-1000/T} = 10^{16} e^{-2000/T}$
 $\frac{e^{-2000/T}}{e^{-1000/T}} = \frac{10^{15}}{10^{16}}$
 $e^{-1000/T} = 10^{-1}$
 $\log_e e^{-1000/T} = \log_e 10^{-1}$
 $\frac{-1000}{T} = 2.303 \log_{10} 10^{-1} = -2.303$
 $T = \frac{1000}{2.303} \text{ K}$
[New NCERT Class 12th Page No. 80]

60. (3)
 - Colliding molecules must have sufficient energy (threshold energy) for effective collision.
 - There should be proper orientation of colliding molecules.
[New NCERT Class 12th Page No. 83]

61. (1)
From the observation, we can see that changing B has no effect on the rate, and doubling the amount of the A doubles the rate of the reaction. Thus, Rate = $k[A]^1$ satisfies the given data. Thus order = 1
[New NCERT Class 12th Page No. 72]

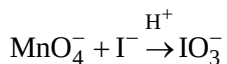
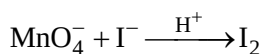
62. (1)
At 298 K, Rate constant (k) of the both of reactions remains the same.
At 60% completion of reaction
 $k = \frac{2.303}{t} \log \frac{a}{a-x}$
 $k = \frac{2.303}{60} \log \frac{100}{100-60} = \frac{2.303}{60} \log \frac{100}{40}$
 $k = \frac{2.303}{60} \log 2.5 = \frac{2.303 \times 0.3979}{60}$
At 50% completion of reaction.
 $t = \frac{2.303}{k} \log \frac{a}{a-x}$
 $t = \frac{2.303 \times 60}{2.303 \times 0.3979} \log \frac{100}{50} \approx 45 \text{ min}$
[New NCERT Class 12th Page No. 73]

63. (1)



[New NCERT Class 12th Page No. 111]

64. (2)



[New NCERT Class 12th Page No. 106]

65. (3)

The actinoids show in general +3 oxidation state. Actinoid contraction is greater from element to element than lanthanoid contraction.

[New NCERT Class 12th Page No. 112]

66. (2)

The first ionisation enthalpies of lanthanoids are around , which is comparable to those of calcium. This similarity in ionization energies is indicative of comparable reactivity and chemical behavior.

[New NCERT Class 12th Page No. 110]

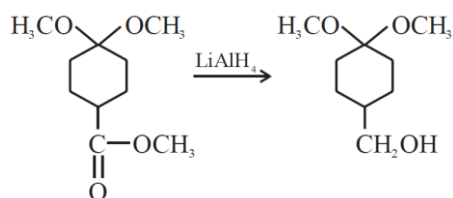
67. (4)

The solubility increases with an increase in the number of branches because the increase in branching increases the surface area of the compounds. Thus, the solubility increases which means that tertiary butyl alcohol will be more soluble in water as compared to n-butyl and isobutyl.

[New NCERT Class 12th Page No. 204]

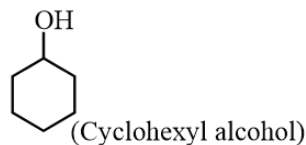
68. (4)

LiAlH_4 reduces esters to alcohols but is inert to ethers.



[New NCERT Class 12th Page No. 200]

69. (1)



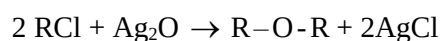
[New NCERT Class 12th Page No. 194]

70. (3)

- A. Hydroboration-oxidation
- B. Oxymercuration-demercuration
- C. Acid catalyzed hydration

[New NCERT Class 12th Page No. 199]

71. (1)



[New NCERT Class 12th Page No. 215]

72. (1)

Ketones are less reactive than aldehydes towards nucleophilic addition reactions.

[New NCERT Class 12th Page No. 236]

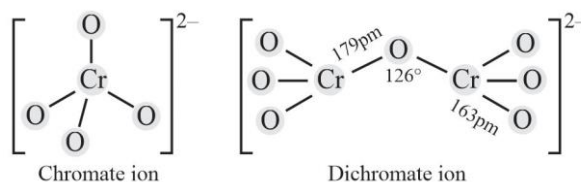
73. (1)

Nitrogen does not have d orbitals to expand its covalence beyond four, hence NCl_3 is possible only. NCl_5 is not possible due to the absence of d-orbital in N. In the case of P and As there is availability of d-orbital, hence PCl_5 and AsCl_5 is possible.

Hence, option (1) is correct.

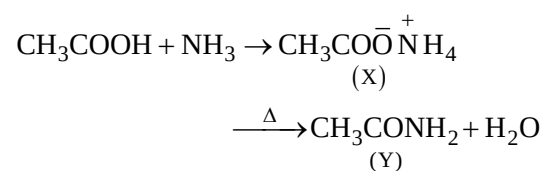
[Newly added topic]

74. (2)



[New NCERT Class 12th Page No. 106]

75. (2)



[New NCERT Class 12th Page No. 241]

76. (4)
Propane-1, 2, 3-tricarboxylic acid
[New NCERT Class 12th Page No. 228]
77. (3)
When aldehydes are heated with ammoniacal silver nitrate solution (Tollen's reagent), it reduces the silver ions to metallic silver and forms a silver mirror. Ketones do not give this test as they are different to oxidise.
[New NCERT Class 12th Page No. 239]
78. (4)
2,4-DNP test is given by aldehydes and ketones.
[New NCERT Class 12th Page No. 237]
79. (3)
Weak intermolecular forces leads to volatility in halogens.
[Newly added topic]
80. (4)
As we move down the group size of metal increases therefore, M—H bond length increases and hence stability decreases. So NH₃ is the most stable hydride.
[Old NCERT Class 12th Page No. 167]
81. (2)
Atomic and ionic radii of group 13 elements are lower than those of alkaline earth metals of group 2 primarily due to greater nuclear charge of group 13 elements as compared to group 2 elements. On moving down the group the atomic radius of gallium is slightly lower than that of Al. This is due to the presence of d electrons in Ga which do not shield the nucleus effectively. As a result, the electrons in gallium experience greater force of attraction by the nucleus than in Al and hence the atomic radius of Ga 135pm is slightly less than that of Al 143pm. Thus, the increasing order of atomic radii of group 13 elements is B < Ga < Al < In < Tl.
[Newly added topic]

82. (4)
The ionisation enthalpy values as expected from the general trends, do not decrease smoothly down the group. The decrease from B to Al is associated with increase in size. The observed discontinuity in the ionisation enthalpy values between Al and Ga, and that between In and Tl are due to the inability of d- and f-electrons, which have low screening effect, to compensate the increase in nuclear charge down the group.
[Newly added topic]
83. (1)
 $\text{Ge}^{2+} > \text{Sn}^{2+} > \text{Pb}^{2+}$
[Newly added topic]
84. (2)
Carbon-60 Contain 12 Pentagons and 20 hexagons.
[Newly added topic]
85. (2)
Boron compounds behave as Lewis acids because of their electron deficient character.
[Newly added topic]
86. (1)
H₂O shows hydrogen bonding and thus has maximum boiling point.
[Newly added topic]
87. (1)
The greater the element-element bond strength, the higher is the tendency towards catenation. As the size increases from S to Te, the bond dissociation energy decreases. O-O bond energy is, however, smaller than that of S-S because of inter electronic repulsions within the smaller size of oxygen atoms.
Thus, sulphur has a maximum tendency towards catenation.
[Newly added topic]

88. (1)
-3, +3 and +5

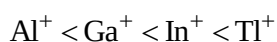
[Newly added topic]

89. (2)

Group 14 element	Oxidation state
C	+4
Si	+4
Ge, Sn, Pb	+2, +4

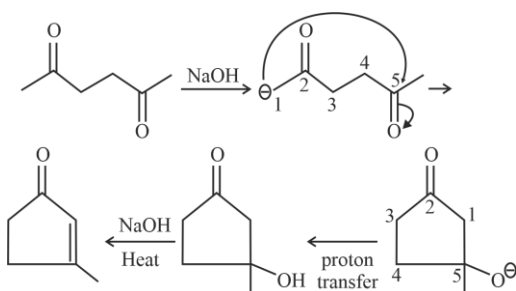
[Newly added topic]

90. (2)



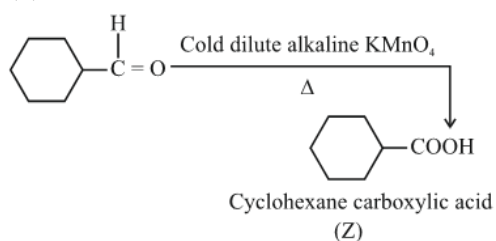
[Newly added topic]

91. (3)



[New NCERT Class 12th Page No. 241]

92. (4)



[New NCERT Class 12th Page No. 239]

93. (4)

For zero order reaction, $[A] = [A_0] - kt$

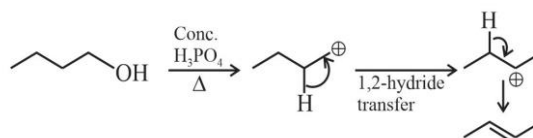
[New NCERT Class 12th Page No. 65]

94. (3)

All radioactive disintegration reactions are examples of first order kinetics.

[New NCERT Class 12th Page No. 75]

95. (3)



[New NCERT Class 12th Page No. 204]

96. (1)

The reactivity of halogen acids (HI, HBr, HCl) with ethanol ($\text{C}_2\text{H}_5\text{OH}$) in the presence of zinc chloride (ZnCl_2) follows the order: $\text{HI} > \text{HBr} > \text{HCl}$. HI is the strongest acid among the three followed by HBr and then HCl. Stronger acids are more readily available to donate protons making them more reactive.

[New NCERT Class 12th Page No. 208]

97. (4)

$\frac{P}{Q}$, PQ , $\frac{dP}{dQ}$ are intensive properties.

[New NCERT Class 11th Page No. 144]

98. (2)

$$\Delta_{\text{vap}}H^\circ = 35 \text{ kJ mol}^{-1} \text{ and } T_{\text{boiling}} = 350 \text{ K}$$

$$\Rightarrow \Delta_{\text{vap}}S^\circ = \frac{\Delta_{\text{vap}}H^\circ}{T_{\text{boiling}}} = 100 \text{ J mol}^{-1} \text{K}^{-1}$$

$$\Delta_{\text{cond}}S^\circ = -100 \text{ J mol}^{-1} / \text{K}^{-1}$$

[New NCERT Class 11th Page No. 159]

99. (2)

$$\text{Ti}_{(22)} = 3d^2 4s^2; \text{Ti}^{2+} = 3d^2$$

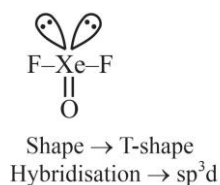
$$\text{V}_{(23)} = 3d^3 4s^2; \text{V}^{3+} = 3d^2$$

$$\text{Cr}_{(24)} = 3d^4 4s^2; \text{Cr}^{4+} = 3d^2$$

$$\text{Mn}_{(25)} = 3d^5 4s^2; \text{Mn}^{5+} = 3d^2$$

[New NCERT Class 12th Page No. 102]

100. (4)



[Newly added topic]

SECTION-(III) BOTANY

101. (4)

Most common and prevalent Mendelian disorders are Haemophilia, Cystic fibrosis, Sickle-cell anaemia, Colour blindness, Phenylketonuria, Thalassemia, etc. Down's syndrome, Turner's syndrome, Klinefelter's syndrome are common examples of chromosomal disorders.

[New NCERT Class 12th Page No. 73]

102. (1)

The statement "Polygenic character is controlled by multiple alleles" is not correct.

Here's why:

Polygenic characters are controlled by multiple genes, not just multiple alleles. These genes typically have additive effects, and each gene contributes to the phenotype. Examples include traits like height, skin color, and intelligence in humans.

[New NCERT Class 12th Page No. 69]

103. (2)

Heterozygous axial flower – Aa

Terminal flower – aa

$$\text{Axial flower} = \frac{2}{4} \times 100 = 50\%$$

$$\text{Terminal flower} = \frac{2}{4} \times 100 = 50\%$$

[New NCERT Class 12th Page No. 57]

104. (4)

If a genetic disease is transferred from a phenotypically normal but carrier female to only some of the male progeny, the disease will be sex-linked recessive.

[New NCERT Class 12th Page No. 75]

105. (3)

In a test cross of dominant phenotype pea plant having a violet color flower with recessive parent result in violet and white flower, then genotype for test cross is- Ww × ww.

In a test cross, the cross is done with recessive genotype.

According to the given question, the genotype of the violet color flower will be either WW or Ww. But its resultant offspring shows both violet and white flower, so it must be heterozygous.

So the genotype will be Ww × ww.

[New NCERT Class 12th Page No. 58]

106. (3)

Person affected with phenylketonuria lack an enzyme that converts amino acid phenylalanine into tyrosine.

[New NCERT Class 12th Page No. 75]

107. (1)

Stem height	Dwarf
Flower colour	Violet
Pod shape	Constricted
Seed colour	Green

[New NCERT Class 12th Page No. 55]

108. (2)

In co-dominance, both alleles of a gene are independently expressed in the heterozygote, leading to a phenotype that combines features of both alleles. Neither allele is dominant or recessive; instead, they contribute equally to the observed traits.

[New NCERT Class 12th Page No. 61]

109. (2)

Turner syndrome is a chromosomal disorder in which a female is borne with only one X chromosome i.e 45 chromosomes (XO) condition. Such females have rudimentary ovaries becomes sterile.

[New NCERT Class 12th Page No. 76]

110. (4)

Colourblind sons and carrier daughter.

[New NCERT Class 12th Page No. 73]

111. (3)

If two gene experience independent assortment, the gene are located on different chromosomes or are far apart on the same chromosome.

[New NCERT Class 12th Page No. 64]

112. (2)

XO type of sex chromosomes determine male sex in grasshoppers.

[New NCERT Class 12th Page No. 76]

113. (3)

If an individual with genotype TtRr is crossed with similar genotype and if both gene T and R are linked and crossing over not occur then probable genotype is TtRr, ttrr, TTRR.

[New NCERT Class 12th page No. 63, 64]

114. (3)

ABO blood groups are controlled by the gene *I*. The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene.

[New NCERT Class 12th Page No. 61]

115. (3)

Punnett square is a graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross. Punnett square is used to understand production of gametes by the parents.

[New NCERT Class 12th Page No. 63]

116. (2)

The domestic sewage in large cities is processed by aerobic and then anaerobic bacteria in the secondary treatment in Sewage Treatment Plants (STPs).

[New NCERT Class 12th Page No. 154]

117. (4)

Citric acid	<i>Aspergillus niger</i>
Baker's yeast	<i>Saccharomyces cerevisiae</i>
Statins	<i>Monascus purpureus</i>
Biocontrol	<i>Nucleopolyhedrovirus</i>

[New NCERT Class 12th Page No. 152, 153]

118. (3)

Cyclosporin A prevent the body from rejecting the donor tissue, immunosuppression can be purposefully induced with medication, such as in the case of bone marrow or other organ transplantation.

Cyclosporin is produced by *Trichoderma polysporum*.

[New NCERT Class 12th Page No. 153]

119. (2)

Bacillus thuringiensis.

[New NCERT Class 12th Page No. 153]

120. (1)

While working on *Staphylococci* bacteria, Alexander Fleming noticed a mould growing in one of his unwashed culture plates around which *Staphylococci* could not grow. He discovered that it was caused by a chemical produced by the mould and named it Penicillin after the mould *Penicillium notatum*.

[New NCERT Class 12th Page No. 152]

121. (2)

Primary treatment steps basically involve physical removal of particles - large and small - from the sewage through filtration and sedimentation.

[New NCERT Class 12th Pg. No. 154]

122. (2)

Big holes in Swiss cheese are formed as a result of a bacterium producing a large amount of carbon dioxide.

[New NCERT Class 12th Page No.151]

123. (2)

Saccharomyces cerevisiae is used in the production of beverages.

[New NCERT Class 12th Page No. 152]

124. (4)

BOD of wastewater is estimated by measuring the amount of oxygen consumption.

[New NCERT Class 12th Page No. 154]

125. (4)

Toddy is made by fermenting sap from palm trees. Microbes are also used to ferment fish, soyabean and bamboo-shoots. Microbes are also used for making cheese.

[New NCERT Class 12 Page No. 151]

126. (3)

Baculoviruses are excellent candidates for species specific, narrow spectrum insecticidal applications.

[New NCERT Class 12th Page No. 157]

127. (3)

A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B_{12} .

[New NCERT Class 12th Page No. 151]

128. (4)

Blue green algae	Add organic matter to soil
IPM	Integrated pest management
<i>Trichoderma</i>	Present in root ecosystems
Dragonflies	Get rid of mosquitoes

[New NCERT Class 12th Page No. 157, 158]

129. (4)

Silk dewaxing and degumming have been accomplished using a lipase enzyme in conjunction with a protease enzyme.

[New NCERT Class 11th Page No. 153]

130. (4)

Trichoderma can be used as a biocontrol agent in the treatment of plant disease.

[New NCERT Class 12th Page No. 157]

131. (2)

Cheese and yogurt are products of fermentation. During fermentation, microorganisms like bacteria and yeast act on milk components, converting lactose into lactic acid. This process enhances flavor, texture, and nutritional value, contributing to the characteristic properties of cheese and yogurt.

[New NCERT Class 12th Page No. 151]

132. (1)

In an angiospermic plant the embryo develop at micropylar end of embryo sac.

[New NCERT Class 12th Page No. 9]

133. (1)

A true fruit develops from the ovary.

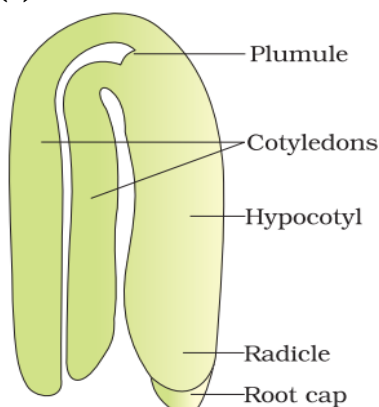
[New NCERT Class 12th Page No. 20]

134. (4)

Hybrid varieties of several of our food and vegetable crops are being extensively cultivated. Cultivation of hybrids has tremendously increased productivity. One of the problems of hybrids is that hybrid seeds have to be produced every year. Production of hybrid seeds is costly.

[New NCERT Class 12th Page No. 23]

135. (2)



[New NCERT Class 12th Page No. 19]

136. (2)

In some dicots, no distinct endosperm can be seen because the endosperm is absorbed early during seed development, or it is used up completely before the seed matures. This can happen in plants where the embryo develops rapidly and the nutrients stored in the endosperm are transferred to the growing embryo.

[New NCERT Class 12th Page No. 19]

137. (4)

1. Three haploid antipodal cells are present.
2. In the embryo sac, the antipodal cells are found at the micropyle's opposite end.
3. The egg cell, which is on the opposite end of the embryo sac, is flanked by two synergid cells.
4. The egg cells are haploid as well.
5. As a result, the number of chromosomes in egg cells and antipodal cells is haploid.

[New NCERT Class 12th Page No. 11]

138. (1)

As ovules mature into seeds, the ovary develops into a fruit, i.e., the transformation of ovules into seeds and ovary into fruit proceeds simultaneously. The wall of the ovary develops into the wall of fruit called pericarp.

[New NCERT Class 12th Page No. 20]

139. (2)

- Autogamy - In this, an intersexual or perfect flower is pollinated by its pollen. It is possible when anther and stigma are close together.
- Geitonogamy - In this, the Pollen grains of a 1 flower are transferred to the stigma of another flower belonging to either the same plant or genetically similar plants. It often shows modifications similar to ones found in xenogamy or cross-pollination.
- Monoecious (*Bisexual*) plant bears either bisexual or unisexual flowers and they can exhibit both autogamy as well as geitonogamy.
- Dioecious(*unisexual*)plants bear only male or female flowers and they will not show autogamy or geitonogamy hence, only xenogamy is possible.

[New NCERT Class 12th Page No. 11,12]

140. (3)

The male gametes of Rice plant have 12 chromosomes in their nucleus.

$n=12$,

So, the female gametes of Rice plant have 12 chromosomes.

$2n=24$,

So, the zygote have 24 chromosomes.

[New NCERT Class 12th Page No.18]

141. (1)

The hard outer layer called the exine is made up of Sporopollenin. The exine exhibits a fascinating array of patterns and designs.

[New NCERT Class 12th Page No. 8]

142. (1)

Following compatible pollination, the pollen grain germinates on the stigma to produce a pollen tube through one of the germ pores.

[New NCERT Class 12th Page No.16]

143. (3)

Each PMC undergoes meiosis to give rise four pollen grains.

[New NCERT Class 12th Page No. 6]

144. (2)

Adaptation shown by pollen grain by hydrophilous flower is mucilage coat on pollen grains.

[New NCERT Class 12th Page No.13]

145. (2)

In *Vallisneria*, the female flower reaches the surface of water by the long stalk and the male flowers or pollen grains are released onto the surface of water.

In *Yucca* plants, pollination takes place by the help of moth insects.

Mucilaginous covering on pollen grains is seen in water pollinated plants and Maize belong to monocots or grass family which are pollinated by wind.

[New NCERT Class 12th Page No. 13]

146. (3)

Endosperm development precedes embryo development, ensuring that the embryo has the necessary resources for its development.

[New NCERT Class 12th Page No. 18, 19]

147. (3)

The portion of embryonal axis above the level of cotyledons is the epicotyl.

[New NCERT Class 12th Page No. 19]

148. (3)

Polyembryony	Occurrence of more than one embryo in a seed.
Apomixis	Seed formation without fertilisation
Dormancy	Inactive state of embryo
False fruit	Thalamus also contributes to fruit formation.

[New NCERT Class 12th Page No. 20, 22 and 23]

149. (3)

Bagging female flower buds in plants with unisexual flowers is necessary to prevent unwanted pollination from external pollen sources, ensuring controlled breeding and study. Bisexual flowers exhibiting protogyny can be used for controlled pollination if properly managed.

[New NCERT Class 12th Page No. 17]

150. (4)

Tassels of a corn cob	Wind pollination
Beetles secrete foul odours	Insect pollination
<i>Yucca</i>	Get pollinated by moth
<i>Amorphophallus</i>	Provide safe place to lay eggs for insect

[New NCERT Class 12th Page No. 12, 13, 14]

SECTION-(IV) ZOOLOGY**151. (2)**

Decreasing the number of people reaching reproductive age would not increase the population.

[New NCERT Class 12th Page No. 43]

152. (4)

- Both male and female can be infertile.
- Infertile couples can have children with the help of ARTs.

[New NCERT Class 12th Page No. 47, 48]

153. (4)

- The diseases which are transmitted through sexual intercourse are termed sexually transmitted diseases (STDs). Examples are syphilis, AIDS, chlamydia, herpes & gonorrhoea, etc.
- Early symptoms of STDs include:
 - Painless sores
 - Slight pain and swelling
 - Fluid discharge from the vagina
 - Itching in the genital region

[New NCERT Class 12th Page No. 47]

154. (2)

The primary goal of implementing action plans for reproductive health is to provide medical assistance and care for issues related to reproduction, such as pregnancy, STDs, infertility, etc.

[New NCERT Class 12th Page No. 46]

155. (2)

MTPs are safe during the first trimester.

[New NCERT Class 12th Page No. 46]

156. (2)

Figure: Condom for female

- Condom is the barrier method, ovums and sperms are prevented from physically meeting with the help of this barrier.
- Condoms protect from sexual disease like STD and AIDS.

[New NCERT Class 12th Page No. 44]

157. (1)

- Genital herpes is venereal disease which is not curable even if it is detected early and treated properly.
- In vasectomy, a small part of the vas deferens human of male is tied or cut.

[New NCERT Class 12 Page No. 47]

158. (2)

Childless couples can have children through a type of assisted reproductive technology called GIFT (gamete intra fello-pian transfer) where ovum will be transferred from the female donor into fello-pian tube of the female who wants baby.

[New NCERT Class 12th Page No. 48]

159. (3)

‘Saheli’ is specifically described as a new oral contraceptive for females, indicating its primary function.

[New NCERT Class 12th Page No. 43]

160. (2)

- The female gamete production starts in the embryonic stage, that is even before the birth.
- Gonadotropin releasing hormone controls the process of oogenesis at the time of puberty.

[New NCERT Class 12th Page No. 32]

161. (2)

- The extinct human ancestor who ate only fruits and hunted with stone weapons was *Australopithecus*.
- They lived in East African grasslands about 2 million years ago.
- They were about 1.5 m high and their brain capacity was about 300-500 c.c. They had bipedal locomotion and erect posture.

[New NCERT Class 12th Page No. 124]

162. (4)

Vestigial structures - The structures that are present in reduced form but were properly functional in our ancestors. They got reduced due to the course of evolution.

Examples - Vermiform appendix, Caudal vertebrae (coccyx/tail), Body hair, Third molar or Wisdom tooth.

Epiglottis is a flap-like structure present to cover the glottis and it is not a vestigial organ.

[New NCERT Class 12th Page No. 113]

163. (3)
Ernst Haeckel proposed embryological support for evolution.

[New NCERT Class 12th Page No. 113]

164. (3)
Louis Pasteur disapproved the theory of spontaneous generation.

[New NCERT Class 12th Page No. 111]

165. (3)

Barrier method	Cervical cap
Natural method	Coitus interruptus
IUDs	Copper-T
Subcutaneous method	Implant

[New NCERT Class 12th Page No. 44]

166. (3)
Coelacanth are thought to be evolved into first amphibians.

[New NCERT Class 12th Page No. 122]

167. (1)
Amniocentesis is the procedure to detect chromosomal pattern of the foetus by obtaining amniotic fluid.

[New NCERT Class 12th Page No. 42]

168. (1)
Vaults is a type of barrier method.

[New NCERT Class 12th Page No. 44]

169. (1)
When we consider the sterilisation method then the following observation takes place.

- Sterilisation method in the male is called vasectomy.
- Sterilisation in the female is called Tubectomy.
- In case of vasectomy, a small part of the vas deferens is cut and tied up through a small incision on the scrotum that prevent the transport of sperm from testis to the urethra.
- Sterilisation is highly effective but there reversibility is very poor.
- Therefore, Sterilisation is a procedure that renders an individual incapable of further reproduction.

[New NCERT Class 12th Page No. 45]

170. (4)
From his experiments, S.L. Miller produced amino acids by mixing the CH_4 , H_2 , NH_3 and water vapor at $800^\circ C$ in a closed flask.

[New NCERT Class 12th Page No. 122]

171. (3)
Frequency of heterozygotes = $2pq$
where $p = A = 0.3$
 $q = a = 0.7$

$$\therefore 2pq = 2 \times 0.3 \times 0.7 = 0.42$$

[New NCERT Class 12th Page No. 121]

172. (2)
Flying squirrel and flying phalanger show convergent evolution.

[New NCERT Class 12th Page No. 118]

173. (4)
A gene pool is total genes and their alleles in a population.

[New NCERT Class 12th Page No. 120]

174. (2)
- The meiotic division of the secondary oocyte is arrested at the Metaphase II of the meiosis.
 - The division is completed at the time of fertilisation of the ovum by the sperm.

[New NCERT Class 12th Page No. 33]

175. (2)
- *Homo habilis* probably did not eat meat.
 - *Ramapithecus* was more man-like while *Dryopithecus* was more ape-like.
 - *Homo erectus* had a large brain around 900cc.

[New NCERT Class 12th Page No. 124]

176. (1)
- The uterine lining is constructed by the estrogen hormone during the proliferative phase so that it can develop or proliferate. At the same time that the lining of the uterus regenerates during the Proliferative phase, the primary follicles in the ovary grow.
 - Hence, Change in the female secondary sex organs are controlled by oestrogen of the ovary in the proliferative phase of the menstrual cycle whereas the changes in the secretory phase are controlled by the progesterone hormone.

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177. (4)

Prostate gland is an unpaired gland in the male reproductive system.

178. (3)

One secondary spermatocyte gives rise to two sperms.

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179. (3)

The glandular tissue of each breast is divided into 15-20 mammary lobes.

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180. (1)

The frequency of AA individuals in a population is simply p^2 .

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181. (4)

Placenta acts as an endocrine tissue and produces several hormones like human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens, progestogens, etc.

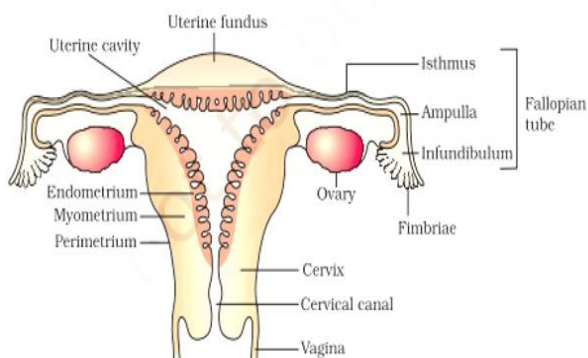
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182. (3)

Colostrum is the first milk produced after parturition.

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183. (2)



II: Ampulla, wider part of oviduct where fertilisation occurs.

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184. (2)

- Implantation occurs after approximately 10 days of fertilization
- Implantation results in increase in the size of uterus and thickening of endometrium.

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185. (2)

- The labia majora are fleshy folds of tissue, which extend down from the mons pubis and surround the vaginal opening.
- The labia minora are paired folds of tissue under the labia majora.
- The opening of the vagina is often covered partially by a membrane called hymen.

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186. (4)

- French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs.
- He gave the examples of Giraffes who in an attempt to forage leaves on tall trees had to adapt by elongation of their necks.
- As they passed on this acquired character of elongated neck to succeeding generations. Giraffes, slowly, over the years, came to acquire long- necks. Nobody believes this conjecture any more.

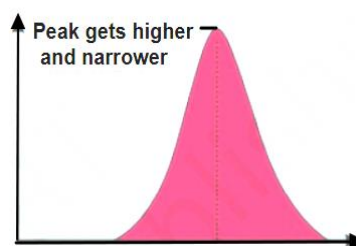
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187. (1)

Gametogenesis	Formation of gametes.
Insemination	Transfer of sperms into female genital tract.
Parturition	Delivery of the baby.
Ovulation	Release of secondary oocyte from ovary.

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188. (1)



It is stabilising selection.

[New NCERT Class 12th Page No. 120]

189. (3)

- The secretions of the acrosome help the sperm enter into the cytoplasm of the ovum through the zona pellucida and the plasma membrane.
- This induces the completion of the meiotic division of the secondary oocyte. The second meiotic division is also unequal and results in the formation of a second polar body and a haploid ovum (ootid).

[New NCERT Class 12th Page No. 32]

190. (1)
The phase of embryonic development during pregnancy is gestation.

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191. (2)

Origin of universe	20 bya
Origin of earth	4.5 bya
Appearance of life after the formation of earth	500 mya
First cellular form of life	2000 mya

[New NCERT Class 12th Page No. 111]

192. (3)

- The atmosphere of primitive earth was reducing in nature because no free oxygen was available at that time.
- *Tyrannosaurus rex* had huge fearsome dagger-like teeth.

[New NCERT Class 12th Page No. 111, 124]

193. (1)

LH surge occurs on 14th day of menstrual cycle.

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194. (4)

According to Darwin:

- Individuals within a population vary in the characteristics they possess.
- Evolution is best viewed as a purposeful and directed change over time.
- Natural selection is the mechanism by which biological evolution takes place.

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195. (2)

By the end of the second month, the foetus develops limbs and digits.

[New NCERT Class 12th Page No. 38]

196. (1)

Cervical canal along with vagina forms the birth canal.

197. (3)

- In luteal phase, remaining parts of the Graafian follicle transform as the corpus luteum.
- The corpus luteum secretes large amount of progesterone which is essential for maintenance of endometrium. Such an endometrium is necessary for implantation of the fertilised ovum and other events of pregnancy.

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198. (3)

Condom is a barrier method that also helps in protecting the user from contracting STDs and AIDS.

[New NCERT Class 12th Page No. 44]

199. (3)

Seminal plasma is formed by the secretion of seminal vesicles, prostate gland and bulbourethral gland.

[New NCERT Class 12th Page No. 28]

200. (3)

- Darwin's finches represent an excellent example of natural selection.
- Natural selection is the process through which organisms adapt and change according to the environment.
- Darwin deduced that these variations made the finches better adapted to take advantage of the food in their particular local environment.
- This arises due to different feeding habits and habitat and is an example of adaptive radiation.

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